

Top 5 Crimes Around Northeastern in 2024

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Introduction

- **Objective:** Analyze crime patterns near Northeastern to better understand crime frequency and trends
- **Insights & Impacts:** Findings can help students, NUPD, and policymakers improve safety measures by identifying peak crime periods and high-risk areas

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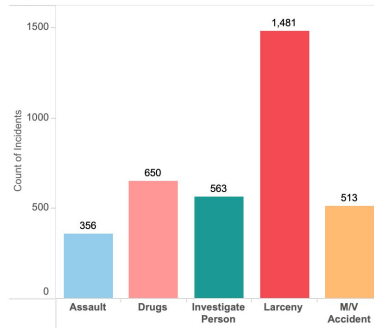


Data Preparation

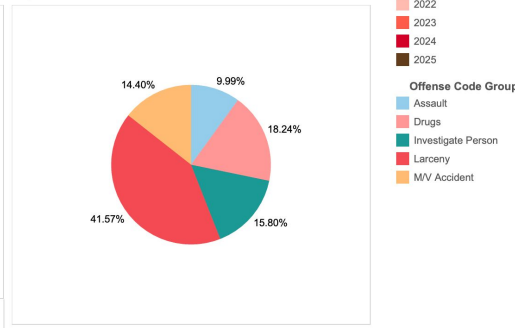
- 1,564,176 missing values
 - Dropped irrelevant columns
 - Replaced missing shooting values
 - Removed unnecessary entries in district column
 - Eliminated missing latitude/longitude values
 - Retained street column missing values
 - Replaced missing offense code group values with corresponding descriptions
- Consolidated similar crimes in the offense code group
- Filtered out incidents from districts B2 and D4
- Created a 1-mile buffer around Northeastern
- Converted date column to datetime format

Types of Crimes

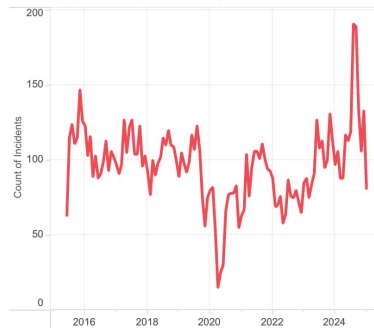
Top 5 Crimes in 2024



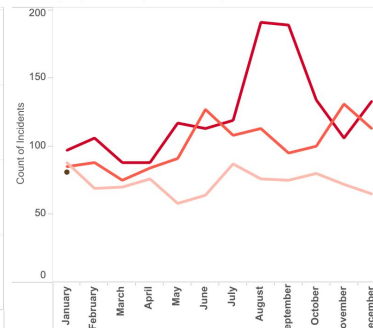
Top 5 Crimes in 2024



Larceny by Month (2022-2025)



Larceny by Month (2022-2025)



- Larceny leads with 1,481 incidents, followed by drug related crimes and investigations
- Larceny accounts for 41.57% of the top 5 crimes
- Larceny peaks around August, indicating seasonal trends
- Understanding these crime trends can help with resource allocation and crime prevention strategies

How can we effectively reduce student risk during peak crime hours?

Offense Code Gr.	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Assault	53	51	52	74	68	46	57	55	56	68	62	73
Drugs	119	108	119	89	109	78	74	77	86	88	99	110
Investigate Person	68	74	67	68	70	79	92	88	65	85	89	105
Larceny	182	194	163	172	208	240	227	304	284	234	237	246
M/V Accident	62	52	61	66	89	60	76	72	70	66	77	75

- Assault
- M/V Accident
- Investigate Person
- Drugs
- Larceny

Day	Red	Orange	Green	Purple	Blue
Sunday	192	76	78	58	54
Monday	221	91	84	72	48
Tuesday	205	116	84	90	46
Wednesday	210	104	84	84	48
Thursday	221	96	77	73	54
Friday	203	97	79	59	60
Saturday	229	70	77	77	46

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Monthly Trends

Top 5 Offenses by Month in 2024

Offense Code Group	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
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- Larceny Surge in August-September:
 - 304 cases in September, 284 in August
 - Strong correlation with student move-in periods
- Winter Drug-Related Incidents:
 - Noticeable increase in drug-related cases during colder month.

Actionable Insights

- Deploy additional security during student move-in periods
- Adjust patrol schedules to address seasonal variations
- Prioritize resource allocation during winter months for drug prevention



Weekly & Daily Distribution

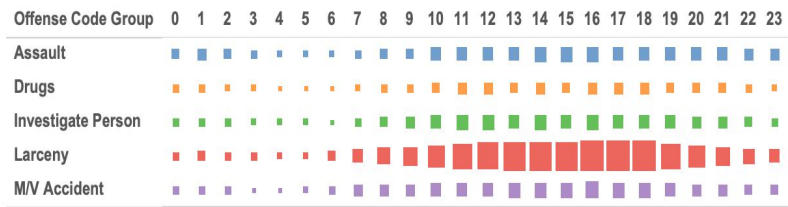
Weekly Patterns

- Tuesday as Peak Crime Day
- Weekend Crime Characteristics:
 - Lower drug activity but higher larceny cases

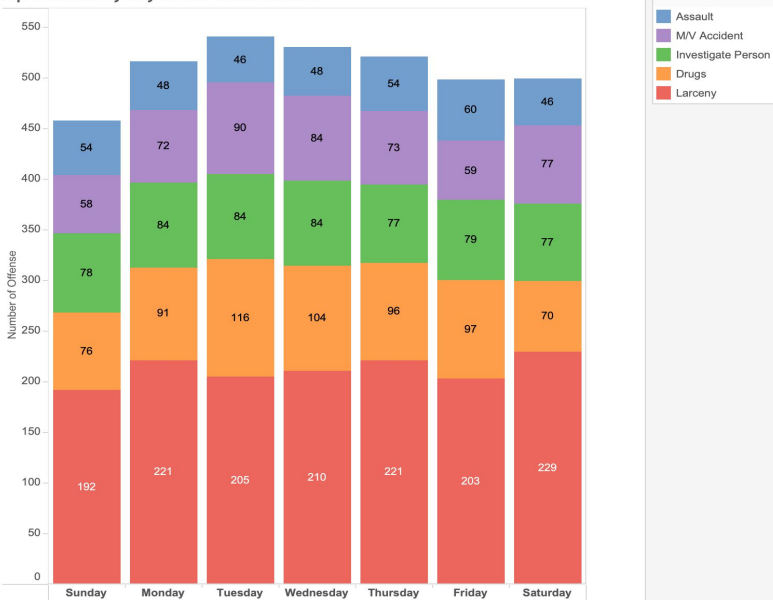
Daily Time Analysis

- High-Risk Hours: 4 PM - 6 PM

Top 5 Crimes by Hour in 2024



Top 5 Crimes by Day of the Week in 2024



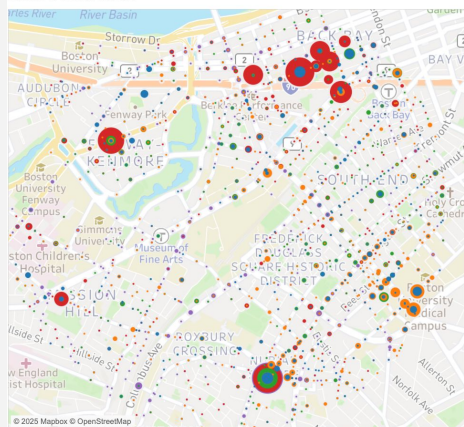
Actionable Insights

- Focus on peak hours (4 PM - 6 PM) for top five crimes
- Allocate more resources during high-risk days like Tuesday



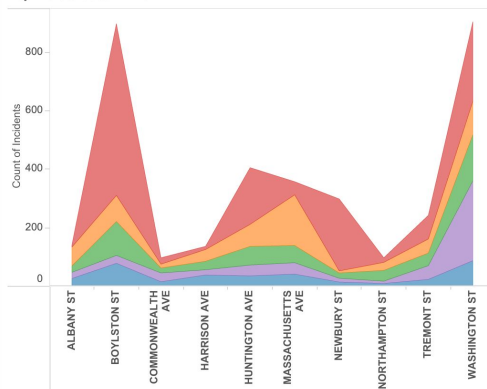
Crime Hotspots

Top 5 Crimes around Northeastern in 2024
Within a 1 mile radius

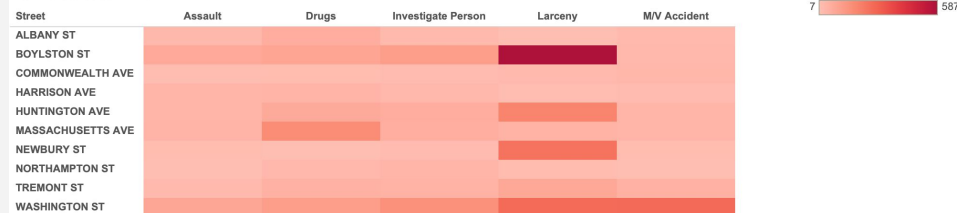


Offense Code Group
Assault Drugs Investigate Person Larceny M/V Accident

Top 10 Streets in 2024



Top 10 Streets in 2024



How can students take precautions, and how can NUPD improve safety measures in areas with higher crime rates around campus?

- **Hotspots:** subway stations, main streets, Back Bay, Fenway
- **Boylston St:** 587 Larceny incidents
- **Washington St:** 272 Larceny incidents, 273 M/V incidents
- **Newbury St:** 246 Larceny incidents
- **Huntington St:** 149 Larceny incidents
- **Massachusetts Ave:** 173 Drug incidents

Conclusion

Our analysis focuses on campus crime patterns, transforming raw data into actionable insights to enhance student safety.

Using three dashboards, we provided a comprehensive view of campus crime by showcasing the top five most common crimes, peak crime periods, and regional crime patterns. By using Tableau, we turned complex crime data into clear, meaningful insights that can help students and NUPD implement preventive measures and improve campus safety.

